

TFT LCD MODULE

2.8 inch 65K 240RGB*320DOTS

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1. LCM Specification

1.1 Description

It is a transmissive type color active matrix liquid crystal display(LCD) which uses amorphous thin film transistor(TFT) as switching devices. This product is composed of a TFT LCD panel, a drive IC, a FPC, a LED-backlight unit. The active display area is 2.8 inches diagonally measured and the native resolution is 240*RGB*320. Features of this product are listed in the following table.

1.2 Functions & Features

Table 1.1 Module Functions & Features

Parameter	Value	Unit
LCD Mode	TFT/Transmissive	-
Color Depth	16.7M	-
Display Resolution	240RGB*320	pixels
Module Size	50.00(H)*69.20(W)*3.88 (T)	mm
Active Area (A.A)	43.20(H)*57.60(W)	mm
Pixel Arrangement	RGB-stripe	-
Viewing Direction	ALL	
Display Mode	Normally Black	
LCD Controller/Driver	ST7789P3	-
IC Package Type	COG	-
Interface	SPI 3W2I Interface	-
Power Supply Voltage	2.8~3.3	V
Back-light	White LED*4	PCS

Technical drawing of the HS-CT28QV-A082 monitor, showing dimensions and components.

Dimensions (mm):

- Top bezel: 50.00 ± 0.2 LCM/TP, 46.20 ± 0.1 LCD OD, 43.60 TP VA, 43.20 LCD AA
- Right bezel: 1.80, 3.20, 3.40
- Left bezel: 69.20 ± 0.2 LCM/TP, 63.88 ± 0.1 LCD OD, 58.00 TP OD, 57.60 LCD AA
- Bottom bezel: 45.50 ± 0.5
- Bottom bezel detail: 3.00, 5.00

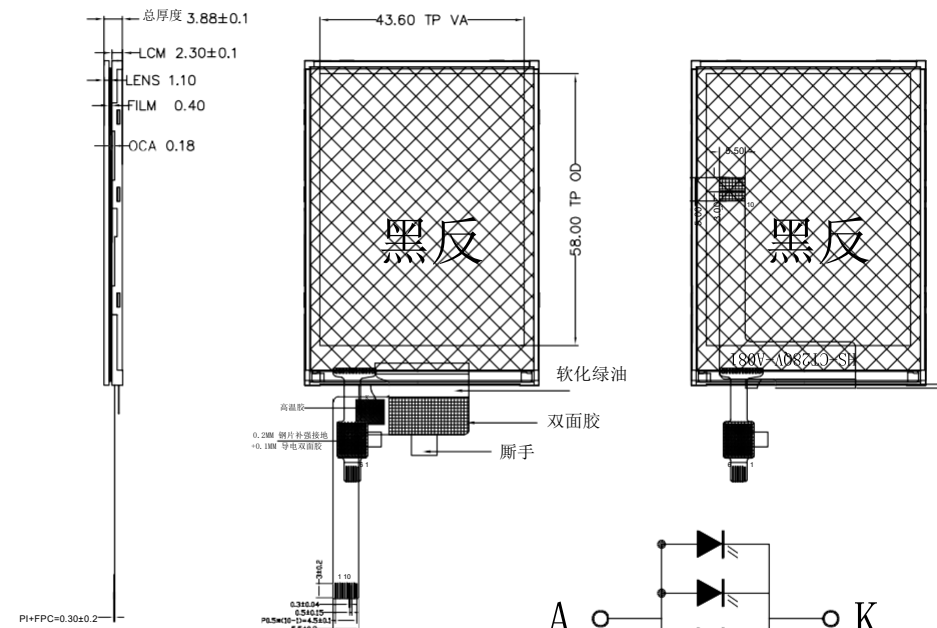
Internal Labels:

- 240*RGB*320
- 2.8 inch TFT
- Viewing Direction ALL
- 黑色马拉胶 TO 03
- HS-CT28QV-A082

Component Label:

- 元器件区域 (Component Area)

1. Display Type: 2.8" _QVGA, Transmissive
2. Driver IC: ST7789P3
3. Viewing Direction: ALL
4. Operating Temperature: -20° - +70°
5. Storage Temperature: -30° - +80°
6. Backlight Type: 4 White LEDs 4并
7. BL Input Current: 80mA/3.0~3.2
8. LCM Brightness: 300 cd/m2 Typ



REV	DESCRIPTION:	DATE
A	初版	2026-05-28

PIN	DESCRIPTION
1	GND
2	RS
3	CS
4	SCL
5	SDA
6	RESET
7	VCC
8	GND
9	LEDA
10	LEDK
PIN	DESCRIPTION
1	TP VDD3.3V
2	TP SCL
3	TP SDA
4	TP INT
5	TP RST
6	GND

FIELD	TOLERANCES					MODULE NO.				MATERIAL		WEIGHT			
L<5	± 0. 05														
5<L<16	± 0. 10					DESIGNED				CHECKED	VERIFIED	APPROVED	FILE NAME		
16<L<63	± 0. 15														
63<L<100	± 0. 20					UNIT	SIZE	SCALE	MOD						
100<L	± 0. 25	mm	A4	1:1											
ANGLE	± 1 °														

3. Pin Descriptions

Pin No.	Symbol	Functional
1	GND	Power Ground
2	RS	SPI data/command selection pin
3	CS	Chip select signal
4	SCL	SPI interface clock input pin.
5	SDA	SPI interface data input/output pin.
6	RESET	LCD Reset signal pin(active low)
7	VCC	Power supply to the interface pins (2.8V)
8	GND	Power Ground
9	LED-A	Backlight , Anode
10	LED-K	Backlight , Cathode

CTP:I2C

No	Name	Description
1	VDD(3.3V)	CTP Power supply for analog
2	SCL-3.3	CTP Serial Clock input pin.
3	SDA-3.3	CTP Serial data input pin.
4	INT-3.3	CTP interrupt pin.
5	RST-3.3	CTP Reset signal pin(activelow)
6	GND	Ground

4. Electrical Units

4.1 Absolute Maximum Ratings

The absolute maximum ratings are list on Table 4.1. When used out of the absolute maximum ratings, the LCM may be permanently damaged. Using the LCM within the following electrical characteristics limit is strongly recommended for normal operation. If these electrical characteristic conditions are exceeded during normal operation, the LCM will malfunction and cause poor reliability.

Table 4.1 Module Absolute Maximum Ratings(VSS=0V)

Item	Symbol	Unit	Value	Note
Power Supply Voltage	VCC	V	-0.3 to +4.5	
Power Supply Voltage	VGH-VSS	V	-0.3 to 19	
Power Supply Voltage	VSS-VGL	V	-0.3 to 19	
Operating Temperature	Top	°C	-20 to +70	
Storage Temperature	Tst	°C	-30 to +80	
Operating Humidity	Hop	%(RH)	10~90	

4.2 Electrical characteristics (Ta=25°C)

Table 4.2:DC Characteristic (Vcc = 2.8 ~ 3.3V)

Item		Symbol	Condition	Min.	Typ.	Max.	Unit
Supply Voltage	Logic	VDD	---	3.0	3.3	3.6	V
Supply Voltage	I/O	IOVCC	---	3.0	3.3	3.6	V
Input Voltage	H level	VIH	---	0.8IOVCC	---	IOVCC	V
	L level	VIL		-0.3	---	0.2IOVCC	
LCD Driving Voltage		VOP	---	---	TBD	---	V

4.3 Back-light Specification

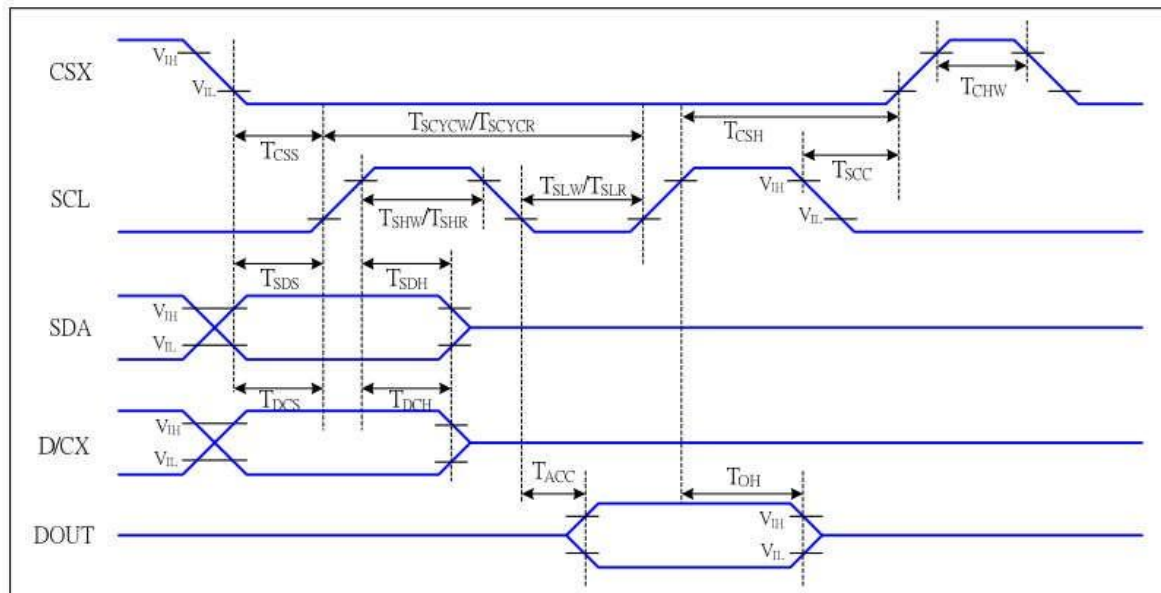
Table 4.3 Back-light Characteristics

Item	Symbol	Conditions	Min.	Typ.	Max.	Unit
Supply Voltage	VF	Only Backlight	2.8	3.0	3.2	V
Supply Current	IF		80			mA
LCM Brightness	IV	IF=80mA	250	300	--	Cd/m2
CIE Color Coordinate	X	IF=80mA	0.25	--	0.31	—
	Y		0.24	--	0.30	
Uniformity	B	IF=80mA	80		—	%
Color	White					

Note: 14 LEDS

5. AC Characteristics

5.1 SPI interface operation



Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	T _{CSS}	Chip select setup time (write)	15		ns	
	T _{CSH}	Chip select hold time (write)	15		ns	
	T _{CSS}	Chip select setup time (read)	60		ns	
	T _{SCC}	Chip select hold time (read)	65		ns	
	T _{CHW}	Chip select "H" pulse width	40		ns	
SCL	T _{SCYCW}	Serial clock cycle (Write)	66		ns	-write command & data ram
	T _{SHW}	SCL "H" pulse width (Write)	15		ns	
	T _{SLW}	SCL "L" pulse width (Write)	15		ns	
	T _{SCYCR}	Serial clock cycle (Read)	150		ns	-read command & data ram
	T _{SHR}	SCL "H" pulse width (Read)	60		ns	
	T _{SLR}	SCL "L" pulse width (Read)	60		ns	
D/CX	T _{DCS}	D/CX setup time	10		ns	
	T _{DCH}	D/CX hold time	10		ns	
SDA (DIN)	T _{SDS}	Data setup time	10		ns	
	T _{SDH}	Data hold time	10		ns	
DOUT	T _{ACC}	Access time	10	50	ns	For maximum CL=30pF
	T _{OH}	Output disable time	15	50	ns	For minimum CL=8pF

6. Optical Specifications

Optical characteristics are determined after the unit has been 'ON' and stable for approximately 30 minutes in a dark environment at 25°C. The values specified are at an approximate distance 50cm from the TFT-LCD surface at a viewing angle of Φ and θ equal to 0°.

Measurement condition: Refer to next pages (C-light source, Halogen Lamp)

*1): with Polarizer *2): without Polarizer *3): Only Color Filter glass

Item	Conditions		Min.	Typ.	Max.	Unit	Note
Viewing Angle (CR>10)	Horizontal	θ_{x+}	35	45	-	degree	(1),(2),(6),(7),(8)
		θ_{x-}	35	45	-		
	Vertical	θ_{y+}	40	50	-		
		θ_{y-}	10	20	-		
Contrast Ratio	Center		400	500	-	-	(1),(3),(6)(7),(8)
Response Time	Rising + Falling		-	16	25	ms	(1),(4),(6),(7),(8)
CF Color Chromaticity (CIE1931)	Red	x	Typ. -0.015	0.626	Typ. +0.015	-	Under C-light
	Red	y		0.336		-	
	Green	x		0.276		-	
	Green	y		0.550		-	
	Blue	x		0.144		-	
	Blue	y		0.130		-	
	White	x		0.307		-	
	White	y		0.329		-	
NTSC	CIE1931		52.5	55.3	-	%	(1),(6),(7),(8)
Transmittance	-		5.8	6.13	-	%	(1),(5),(6),(7),(8)

*Note (1) Definition of Contrast Ratio (CR):

The contrast ratio can be calculated by the following expression.

$$\text{Contrast Ratio (CR)} = L_{255} / L_0$$

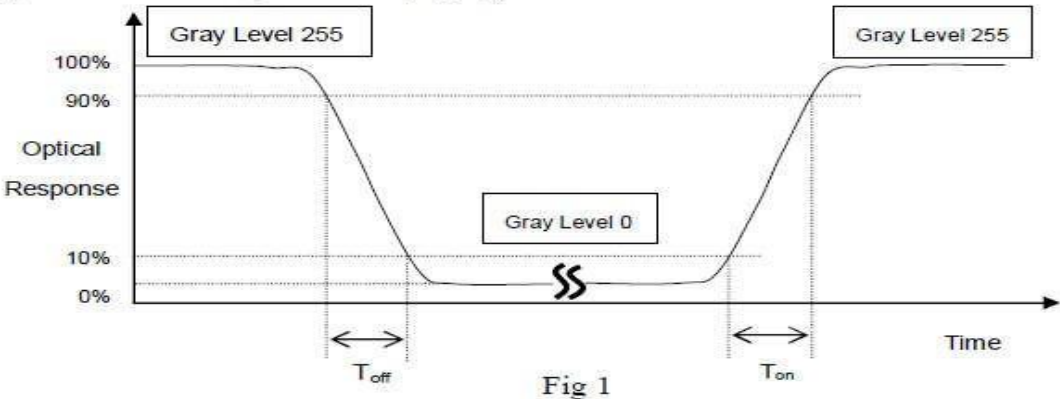
L255: Luminance of gray level 255

L 0: Luminance of gray level 0

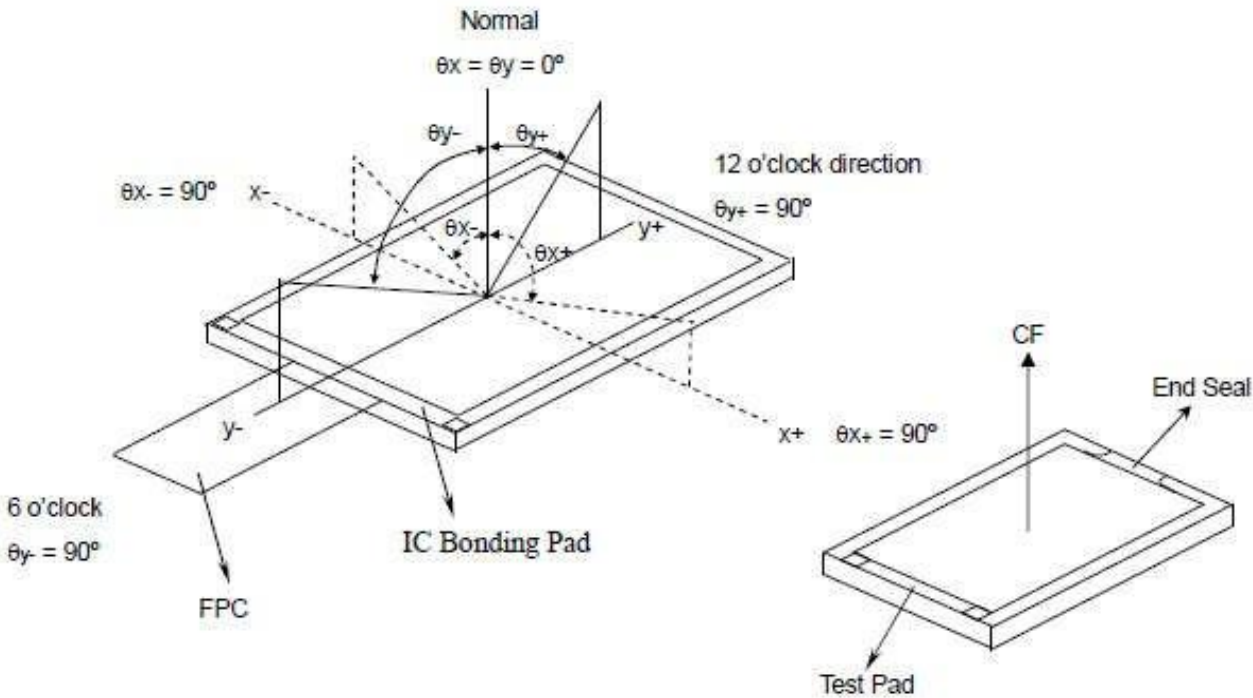
$$CR = CR (5)$$

CR (X) is corresponding to the Contrast Ratio of the point X at Figure in Note (5)

*Note (2) Definition of Response Time (T_{on} , T_{off}):



*Note (3) Definition of Viewing Angle



7. Reliability Test Items

No.	Test Item	Test Condition	Check Time
1	High temp storage	T=80℃	48 hrs
2	Low temp strage	T=-30℃	48 hrs
3	High temp operation	T=70℃	48 hrs
4	Low temp operation	T=-20℃	48 hrs
5	High temp&high humidity	T=60℃ H=90%	48 hrs

Reliability Test Criteria:

Display function should be no change under normal operating condition.

8. Package(TBD)

9. Handling Precautions

9.1 Safety

The liquid crystal in the LCD is poisonous. Keep away from your mouth and eyes. If the liquid crystal contacts with your skin, mouse or clothes, use soap to wash it off immediately.

9.2 Handling

- i. The LCD panel is made by thin glass. Prevent the panel from mechanical shock or putting excessive force on its surface.
- ii. The polarizer attached on the display is very easy to be damaged, handle it with special attention.
- iii. To avoid contamination on the display surface, do not touch the display surface with bare hands.
- iv. The transparent electrodes may be disconnected if you use the LCD panel under dew-condensing environment.
- v. The characteristics of the semiconductor devices may be affected when they are exposed to light, possibly resulting in malfunctioning of the ICs. To prevent such malfunctioning of the ICs, make sure the application and the mounting of the panel are designed so that the IC is not exposed to light.

9.3 Static Electricity

Ground soldering iron tips, tools and testers when you operate. Also ground your body when handling the products and store the products in an anti-electrostatic container.

9.4 Storage

Store the products in a dark place where the temperature is within the range of $25\pm 10^{\circ}\text{C}$ and with low humidity (65%RH or less). Do not store the LCD product in an atmosphere containing organic solvents or corrosive gases.

9.5 Cleaning

Do not wipe the polarizer with dry cloth, as it might cause scratching. Wipe the polarizer with a soft cloth soaked with petroleum IPA. Other chemical might damage the panel.

10. Contact

Contact us if you have any questions, or need customize your display.